

Yukta Seelam

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SUMMARY

Graduate student in Robotics and Autonomous Systems at Boston University with multidisciplinary experience across machine learning, control, and robotic system design. Skilled in developing intelligent, perception-driven robots combining reinforcement learning, computer vision, and hardware integration for real-world autonomy and automation applications.

ACADEMIC PROFILE

Boston University **3.8/4.0**
Masters in Robotics and Autonomous Systems

Expected Jan 2026

Indian Institute of Technology Kanpur
B. Tech in Electrical Engineering

May 2024

EXPERIENCE & RESEARCH

Hybrid Local-Cloud RL for Autonomous Driving in CARLA – Red Hat / Boston University

May 2025 – Sep 2025

- Designed and implemented the UniLCD framework to dynamically switch between local and cloud RL policies, optimizing inference latency by 30 % and reducing compute energy by 22 %.
- Trained PPO-based driving agents in CARLA, integrating multi-sensor inputs (RGB, depth, segmentation) with asynchronous experience replay.
- Built automated logging and visualization pipelines using OpenCV, NumPy, and Matplotlib for real-time rollout evaluation and checkpoint recovery.
- Deployed adaptive offloading based on network bandwidth and policy confidence, improving route-completion reliability in low-connectivity conditions.

Obstacle Avoidance and Path Planning for Rover - IIT Kanpur

Aug 2023 – Mar 2024

- Developed a perception-driven navigation stack for the Clearpath Husky Rover combining LiDAR and stereo-camera data for obstacle detection.
- Fine-tuned YOLOv8 on orchard datasets for fruit and foliage detection, enabling real-time replanning in cluttered fields.
- Integrated ROS Noetic, A*, and Timed Elastic Band planners to compute curvature-bounded trajectories in real time.
- Conducted hardware-in-loop validation achieving a 93 % success rate in field traversal with <10 cm localization error.

Samsung Research & Development – India

May 2023 – Aug 2023

- Deployed deep learning models (YOLOv5, RefineDet) for object detection in robotic vacuum navigation pipelines.
- Enhanced cross-platform model compatibility using TensorRT and ONNX, reducing inference latency by 25%.

Development of AI Combat Robots — IIT Kanpur / ICRA 2022 RoboMaster Challenge

Mar 2022 – Feb 2023

- Engineered two fully autonomous combat robots integrating NVIDIA Jetson GPUs, IMUs, **and** camera arrays for real-time detection and targeting.
- Implemented YOLOv5-based armor-plate classification (8 classes) with 85 % mAP and coupled it with a PID-controlled gimbal for precise shooting under motion.
- Devised visual-marker-based localization using ArUco markers, Kalman filters, and Eigen transform fusion achieving 5 mm positional accuracy.

AWARDS AND PUBLICATIONS

- 3rd Place – IEEE ICRA DJI RoboMaster AI Challenge 2022 (only Indian team internationally)
- Paper: “Towards Autonomous Shooting Rover via Situation-Aware Visual Perception and Dynamic Action Execution”, IEEE/SICE SII 2024 (Vietnam)
- Poster: “Design of AI Ground Robots for Intelligent Combat in a Dynamic Arena”, ICRA 2022

LEADERSHIP POSITIONS

- Team Head – Robotics Club - IITK | President – Toastmasters IIT Kanpur | Captain – IITK Basketball Team | Student Ambassador – BU Robotics

SKILLS & COURSEWORK

- Programming:** Python, C++, MATLAB, Bash, SQL
- ML & AI Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, NumPy, Pandas
- Robotics & Simulation:** ROS, Gazebo, RViz, CARLA, OpenCV, SLAM, Control Systems, Path Planning
- Hardware & Design:** Fusion 360, SolidWorks, Eagle, Arduino, Embedded Systems, 3D Printing, CAD/CAM
- Manufacturing Processes:** Prototyping, Testing, Mechanical Assembly, PCB Design